Multi-Agent Reinforcement Learning for Adversarial Spatial Decision-Making with Temporal Constraints

Chaitanya Chaudhary, Dhruv Meshram, Abhishek Tagalpallewar, Vedant Bondekar
Indian Institute of Technology Gandhinagar
{chaitanya.chaudhary, dhruv.meshram,
abhishek.tagalpallewar, vedant.bondekar}@iitgn.ac.in

September 30, 2026

Abstract

Learning effective policies in adversarial, spatial environments requires agents to adapt to dynamic topologies and temporal resource constraints. We introduce *Modified Wallz*, a turn-based, two-player zero-sum Markov game extending classic wall-placement mechanics with reusable walls, a one-time wall-bypassing hop, and temporary cell blocking. These mechanics force legal moves to change as walls move or cells are blocked. We propose a Multi-Agent Reinforcement Learning (MARL) framework using a Deep Q-Network (DQN) with exact legal-action masking trained via self-play. Policies are evaluated against a hierarchy of non-learning baselines: a spatial shortest-path heuristic, depth-limited minimax search with Alpha-Beta pruning, and an exact retrograde solver for miniature grid configurations. Evaluated across a $2^3 = 8$ ablation matrix, this work investigates how spatial and temporal constraints govern MARL policy stability, trajectory efficiency, and convergence to game-theoretic equilibria.

Keywords: Multi-Agent Reinforcement Learning, Deep Q-Network, Self-Play, Zero-Sum Games, Spatial Planning, Legal-Action Masking

1 Introduction

Sequential decision-making in adversarial spatial domains (such as pursuit-evasion, competitive pathing, and dynamic resource allocation) requires autonomous agents to optimize strategies against actively countering opponents [1, 2]. A central challenge in adversarial navigation arises when environmental topology is governed by state-dependent action spaces and depleting resources. In turn-based spatial games, a single structural placement by an adversary alters the navigable graph for both players, forcing learning algorithms to continuously adapt to non-stationary opponent strategies. Standard grid-world environments often lack tunable mechanics to isolate the effects of transient obstacles and reusable resources on multi-agent learning stability.

To address this gap, we present *Modified Wallz*, a fully observable, sequential zero-sum game built upon classic wall-placement board games. In base Wallz, two pawns race to opposing goal rows on an $N \times N$ grid while placing permanent walls subject to a path-preservation rule preventing total blockade. We extend this foundation with three novel mechanics: (1) *reusable walls* (repositioning a player's most recently placed wall), (2) a *one-time hop* (bypassing an opponent wall or pawn), and (3) *temporary cell blocking* (rendering cells impassable for K plies). Reusable walls force re-evaluation of previously closed paths; the hop introduces a discrete resource dilemma; and temporary blocks induce decaying temporal dependencies. Together, they challenge an agent's ability to plan paths as legal moves change when walls move or cells are blocked. An interactive web prototype is available at: `https://tagalpallewarabhishek.github.io/wallz-custom-game/`.

We investigate three core research questions: (1) How do temporary and reusable spatial constraints impact learned tactical strategies? (2) How does a one-time bypass resource alter long-term planning? (3) How efficiently do value-based MARL policies approximate exact game-theoretic equilibria as board complexity scales? To address these questions, we evaluate a self-play Deep Q-Network (DQN) [3] against a hierarchy of non-learning baselines.

2 Preliminaries

2.1 Markov Games and Zero-Sum Setting

A discrete-time two-player competitive Markov game [2] is defined by $\langle \mathcal{N}, \mathcal{S}, (\mathcal{A}_i)_{i\in\mathcal{N}}, T, (R_i)_{i\in\mathcal{N}}, \gamma \rangle$, where $\mathcal{N} = \{1, 2\}$ is the player set (active player i, opponent $-i = 3 - i$), $\mathcal{S}$ is the state space, $\mathcal{A}_i(s) \subseteq \mathcal{A}_i$ is player i's legal action set in state s, $T : \mathcal{S} \times \mathcal{A}_i \to \mathcal{S}$ is the deterministic transition function ($s_{t+1} = T(s_t, a_t)$), $R_i : \mathcal{S} \times \mathcal{A}_i \times \mathcal{S} \to \mathbb{R}$ is the zero-sum reward function ($R_1 = -R_2$), and $\gamma \in (0, 1]$ is the discount factor. Turn alternation $z_t = z(s_t) \in \{1, 2\}$ designates the acting player at step t.

Player i acts under policy $\pi_i : \mathcal{S} \to \Delta(\mathcal{A}_i)$. The expected return $G_{t,i} = \sum_{k=0}^{\infty} \gamma^k R_i(s_{t+k}, a_{t+k}, s_{t+k+1})$ defines action-value function $Q_i^\pi(s, a) = \mathbb{E}_\pi[G_{t,i} \mid s_t = s, a_t = a]$. In DQN [3], $Q_\theta(s, a)$ is parameterized by neural weights θ. State legality is enforced via a binary legal-action mask [4]:

$$M(s, a) = \mathbb{I}\big(a \in \mathcal{A}_{z(s)}(s)\big), \tag{1}$$

restricting value maximization to legal actions. In self-play [5], agents optimize Q_θ against current or historical policy instances.

2.2 Graph Representation and Spatial Metrics

The navigable board is formalized as an undirected graph $G_t = (V, E_t) = \mathrm{Adj}(C, W_t, \tau_t)$, where $V = C = \{(x, y) \in \mathbb{Z}^2 \mid 1 \le x, y \le N\}$ is the cell vertex set, W_t is the active wall set, and τ_t is the temporary block map. Let $d(u, v; G_t)$ denote geodesic distance on G_t. For pawn position $p_i \in C$ and goal set $G_i \subset C$, $d_i(s) = \min_{g\in G_i} d(p_i, g; G_t)$ denotes shortest-path distance to goal. The binary path-preservation predicate is:

$$\mathrm{PathExists}(s, i) = \mathbb{I}\big(d_i(s) < \infty\big). \tag{2}$$

The relative spatial progress heuristic for player i is defined by shortest-path differential $H_i(s) = d_{-i}(s) - d_i(s)$.

3 Problem Formulation

3.1 Formal Game Model

Modified Wallz is a sequential, deterministic, two-player zero-sum Markov game of perfect information:

$$\mathcal{G} = \big\langle \mathcal{N}, \mathcal{S}, (\mathcal{A}_i)_{i\in\mathcal{N}}, T, (R_i)_{i\in\mathcal{N}}, \gamma, s_0, \mathcal{S}_{\text{terminal}} \big\rangle. \tag{3}$$

Board geometry consists of grid $C = \{1, \ldots, N\}^2$ ($N \ge 3$) and goal rows $G_1 = \{(x, N) \in C\}$, $G_2 = \{(x, 1) \in C\}$. Walls $w = (c, o) \in \mathcal{W}$ (reference cell $c \in C$, orientation $o \in \{\mathrm{H}, \mathrm{V}\}$) block edge set $E(w) \subset E_C$, where E_C is the set of internal cell edges.

3.2 State Representation and Initial State

Environment state $s_t \in \mathcal{S}$ at step $t \in \mathbb{N}_0$ is completely specified by the 10-tuple:

$$s_t = \big(p_1, p_2, W_{1,t}, W_{2,t}, r_1, r_2, h_1, h_2, \tau_t, z_t\big), \tag{4}$$

where $p_1, p_2 \in C$ ($p_1 \ne p_2$) are pawn positions; $W_{i,t} \subset \mathcal{W}$ are disjoint active wall sets ($W_t = W_{1,t} \cup W_{2,t}$, unused wall count $m_i = M - |W_{i,t}|$, $M = 10$); $r_i \in W_{i,t} \cup \{\varnothing\}$ points to player i's currently replaceable wall; $h_i \in \{0, 1\}$ is the hop availability flag; $\tau_t : C \to \{0, \ldots, K\}$ maps cell block durations; and $z_t \in \{1, 2\}$ is the turn indicator. The initial state s_0 is:

$$s_0 = \Big(\big(\lceil N/2 \rceil, 1\big),\ \big(\lceil N/2 \rceil, N\big),\ \emptyset,\ \emptyset,\ \varnothing,\ \varnothing,\ 1,\ 1,\ \mathbf{0},\ 1\Big). \tag{5}$$

3.3 Action Space and Legality Constraints

Active player $i = z_t$ selects legal action $a_t \in \mathcal{A}_i(s_t) = \bigcup_k \mathcal{A}_i^k(s_t)$, partitioned into five legal action categories:

(1) Pawn Move: $a^{\text{move}}(c') \in \mathcal{A}_i^{\text{move}}(s_t) \iff \{p_i, c'\} \in E_C \setminus \bigcup_{w\in W_t} E(w)$, $\tau_t(c') = 0$, $c' \ne p_{-i}$.

(2) Wall Placement: $a^{\text{place}}(w) \in \mathcal{A}_i^{\text{place}}(s_t) \iff |W_{i,t}| < M$, $w \in \mathcal{W} \setminus W_t$, $E(w) \cap \big(\bigcup_{w'\in W_t} E(w')\big) = \emptyset$, $\mathrm{NoCross}(w, W_t) = 1$, $\mathrm{PathExists}(s_t \oplus w, j) = 1, \forall j$.

(3) Wall Replacement: $a^{\text{replace}}(w') \in \mathcal{A}_i^{\text{replace}}(s_t) \iff r_i \neq \varnothing,\ w' \in \mathcal{W} \setminus W_t\,(w' \neq r_i),\ E(w') \cap \Big(\bigcup_{w'' \in W_t \setminus \{r_i\}} E(w'')\Big) = \emptyset,\ \text{NoCross}(w', W_t \setminus \{r_i\}) = 1,\ \text{PathExists}((s_t \ominus r_i) \oplus w', j) = 1, \forall j.$

(4) One-Time Hop: $a^{\text{hop}}(c') \in \mathcal{A}_i^{\text{hop}}(s_t) \iff h_i = 1,\ c' = p_i + 2(c_{\text{inter}} - p_i) \in C$, where $c_{\text{inter}} = p_{-i}$ or $\{p_i, c_{\text{inter}}\} \in E(W_{-i,t})$, with target $\tau_t(c') = 0,\ c' \neq p_{-i}$, no un-hopped wall blocking segment (p_i, c'), and $\text{PathExists}(s_t \xrightarrow{\text{hop}} c', j) = 1, \forall j$.

(5) Temporary Block: $a^{\text{block}}(c^*) \in \mathcal{A}_i^{\text{block}}(s_t) \iff \tau_t(c^*) = 0,\ c^* \notin \{p_1, p_2\},\ \text{PathExists}(s_t \oplus_\tau c^*, j) = 1, \forall j$. Path preservation guarantees $\mathcal{A}_i(s_t) \neq \emptyset$ in all non-terminal states.

3.4 Deterministic Transitions and Terminal Payoffs

State transition $s_{t+1} = T(s_t, a_t)$ updates turn $z_{t+1} = 3 - z_t$ and cell block durations for all $c \in C$: $\tau_{t+1}(c) = K$ if $a_t = a^{\text{block}}(c^*)$ and $c = c^*$, else $\max(0, \tau_t(c) - 1)$. Component updates for acting player $i = z_t$: $a^{\text{move}}(c') \implies p_i' = c';\ a^{\text{place}}(w) \implies W_i' = W_i \cup \{w\},\ r_i' = w;\ a^{\text{replace}}(w') \implies W_i' = (W_i \setminus \{r_i\}) \cup \{w'\},\ r_i' = w';\ a^{\text{hop}}(c') \implies p_i' = c',\ h_i' = 0$.

Terminal states are strictly goal-reaching: $\mathcal{S}_{\text{terminal}} = \mathcal{S}_{1\text{-win}} \cup \mathcal{S}_{2\text{-win}}$, where $\mathcal{S}_{i\text{-win}} = \{s \in \mathcal{S} \mid p_i \in G_i\}$. Outcome space is strictly binary: $\mathcal{O} = \{\text{Player 1 wins}, \text{Player 2 wins}\}$, with no game-theoretic draw states. Zero-sum reward $R_i(s_t, a_t, s_{t+1}) = -R_{-i}(s_t, a_t, s_{t+1})$ yields $+1.0$ if $s_{t+1} \in \mathcal{S}_{i\text{-win}}$, -1.0 if $s_{t+1} \in \mathcal{S}_{-i\text{-win}}$, and 0.0 otherwise. Episodes reaching $T_{\max} = 200$ steps without goal reachability are classified as *experimentally truncated trajectories*. Solution concept corresponds to minimax equilibrium $V_1^*(s) = \max_{\pi_1} \min_{\pi_2} \mathbb{E}_{\pi_1,\pi_2}\left[\sum_{k=0}^{\infty} \gamma^k R_1(s_{t+k}, a_{t+k}, s_{t+k+1}) \mid s_t = s\right]$.

4 Proposed Solutions

We propose a multi-tiered framework comprising three non-learning baselines and a value-based MARL agent trained via self-play.

4.1 Hierarchy of Non-Learning Baselines

Tier 1: Greedy Spatial Heuristic. The heuristic agent selects legal action a^* by strict lexicographical priority: (1) Immediate goal win ($d_i\big(T(s,a)\big) = 0$), (2) Max spatial distance differential $H_i\big(T(s,a)\big)$, (3) Max opponent goal distance d_{-i}, (4) Min pawn goal distance d_i, and (5) Lower action index.

Tier 2: Minimax Search with Alpha-Beta Pruning. For multi-step lookahead $D \in \{2, 4, 6, 8\}$, the minimax value $V_i(s, d)$ recurses over alternating turns with running $[\alpha, \beta]$ bounds. Leaf states are evaluated by $U_i(s) = +1.0$ for win, -1.0 for loss, and $\tanh(0.2 \cdot H_i(s))$ at non-terminal depth $d = 0$. Move ordering prioritizes winning actions, max $H_i(s')$, pawn moves, and wall/block placements, subject to execution limit $t_{\max} = 2.0$ s/turn.

Tier 3: Exact Retrograde Solver. On tractable miniature configurations $\mathcal{C}_{\text{exact}} = \{(N, M, K) \mid N \leq 4, M \leq 2, K \leq 1\}$ ($|\mathcal{S}| \leq 10^6$), an exact solver constructs the finite state-transition graph and performs undiscounted ($\gamma = 1.0$) backward value iteration from goal states $V^*(s) = R_1(s) \in \{+1, -1\}$ [6]:

$$V^*(s) = \begin{cases} \max_{a \in \mathcal{A}_1(s)} V^*\big(T(s,a)\big), & \text{if } z(s) = 1, \\ \min_{a \in \mathcal{A}_2(s)} V^*\big(T(s,a)\big), & \text{if } z(s) = 2. \end{cases} \tag{6}$$

This establishes ground-truth game-theoretic minimax values $V^*(s) \in \{+1, -1\}$ to evaluate RL policy convergence.

4.2 DQN-Based Self-Play Architecture

Canonical State Encoding. State s_t is encoded into spatial tensor $X_t \in \mathbb{R}^{11 \times N \times N}$ and scalar vector $x_t^{\text{scalar}} \in \mathbb{R}^6$ from the active player's canonical perspective ($z(s_t) = 1$). Spatial channels encode: (0) active pawn p_i, (1) opponent pawn p_{-i}, (2) active player horizontal walls $W_{i,t}^{\text{H}}$, (3) active player vertical walls $W_{i,t}^{\text{V}}$, (4) opponent horizontal walls $W_{-i,t}^{\text{H}}$, (5) opponent vertical walls $W_{-i,t}^{\text{V}}$, (6) continuous block duration countdown $\tau_t(c)/K \in [0, 1]$ for Markov completeness, (7) active player goal row G_i, (8) opponent goal row G_{-i}, (9) active replaceable wall indicator r_i, and (10) opponent replaceable wall indicator r_{-i}. Scalar vector $x_t^{\text{scalar}} = [|W_{i,t}|/M, |W_{-i,t}|/M, \mathbb{I}(r_i \neq \varnothing), \mathbb{I}(r_{-i} \neq \varnothing), h_i, h_{-i}]^\top$ captures wall capacities and hop resources.

Action Indexing and Masked Negamax Q-Learning. Actions are mapped into discrete universal space $\mathcal{A}_{\text{enc}}(N)$ of dimension $|\mathcal{A}_{\text{enc}}(N)| = 4(N-1)^2 + N^2 + 8$ ($N = 3 \to 33$, $N = 5 \to 97$). Spatial tensor X_t passes through a 3-layer CNN (32, 64, 64 channels), concatenated with x_t^{scalar}, and processed by a 2-layer MLP (256 units) to output raw Q-values. Exact action masking with $M(s_t, a)$ from Eq. (1) [4] filters invalid choices:

$$a_t = \arg \max_{a: M(s_t,a)=1} Q_\theta(s_t, a). \tag{7}$$

Loss parameters θ minimize negamax temporal-difference error over replay buffer $\mathcal{D}$ [3]:

$$L(\theta) = \mathbb{E}_{(s_t,a_t,r_t,s_{t+1},\text{done})\sim\mathcal{D}} \left[\left(r_t - \gamma(1-\text{done}) \max_{a': M(s_{t+1},a')=1} Q_{\theta^-}(s_{t+1}, a') - Q_\theta(s_t, a_t) \right)^2 \right]. \tag{8}$$

Agents are trained using Adam ($\eta = 3 \times 10^{-4}$, batch size 128, $\gamma = 0.99$) with a 10^4-step heuristic warmup and a self-play pool $\mathcal{P}_{\text{opp}}$ updated every 5,000 steps over 5×10^5 total steps across 5 random seeds.

4.3 Experimental Design and Evaluation Protocol

We evaluate four research hypotheses:

- **H1 (Search Branching):** Reusable walls and temporary blocks expand branching factor $b(s)$ by $\geq 40\%$, reducing Minimax depth under fixed time budget $t_{\max}$.
- **H2 (Sample Efficiency):** DQN self-play with pre-training achieves $\geq 75\%$ win rate against Tier 1 within 2×10^5 steps, versus $\leq 55\%$ for pure self-play.
- **H3 (Equilibrium Convergence):** Mean Q-value error $|Q_\theta(s, a) - Q^*(s, a)|$ on solvable grids ($N \leq 4$) converges below 0.15 overall across training epochs (subject to temporary self-play non-monotonic fluctuations).
- **H4 (Non-Additive Complexity):** Combining all three mechanics increases mean trajectory length beyond individual additive mechanics.

Ablation Matrix and Metrics. Experiments evaluate all $2^3 = 8$ mechanic combinations $(m_1, m_2, m_3) \in \{0, 1\}^3$ across 500 matches per pairing (with initial 2-ply randomized opening states to evaluate non-identical game trajectories between deterministic baselines) using: (1) Win rate P_{win} ($P_{\text{loss}} = 1 - P_{\text{win}}$), (2) Truncation rate P_{trunc} (fraction reaching $T_{\max} = 200$), (3) Mean episode length $\bar{L}$, (4) Optimal Action Agreement $\text{Agr} = \frac{1}{|\mathcal{S}_{\text{eval}}|} \sum_s \mathbb{I}(a_{\text{DQN}}(s) \in \arg\max_a Q^*(s, a))$, and (5) Mean Value Error $E_Q = \frac{1}{|\mathcal{S}_{\text{eval}}|} \sum_s |V_{\text{DQN}}(s) - V^*(s)|$. Technical risk contingencies include potential-based reward shaping $F(s, s') = \gamma H_i(s') - H_i(s)$ [7] for sparse reward stagnation and candidate action pruning for $N \geq 7$.

5 Contribution

Team member execution responsibilities are detailed in Table 1.

Table 1: Team member responsibility matrix.

Member	Assigned Responsibilities
Chaitanya Chaudhary	Minimax search engine (α-β pruning), leaf evaluation $U_i(s)$, move ordering optimization, and factorial ablation experiments ($2^3 = 8$).
Dhruv Meshram	Core Markov game environment engine, action generator $\mathcal{A}_i(s)$, legal action masking, and greedy spatial heuristic baseline.
Abhishek Tagalpallewar	Exact retrograde solver ($\mathcal{C}_{\text{exact}}$), state-transition graph builder, equilibrium metrics (E_Q, Agr), and interactive web prototype.
Vedant Bondekar	Masked DQN neural network architecture (CNN+MLP), self-play training pipeline, replay buffer $\mathcal{D}$, and hyperparameter tuning.

References

[1] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.

[2] Michael L. Littman. Markov games as a framework for multi-agent reinforcement learning. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 157–163, 1994.

[3] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

[4] Shengyi Huang and Santiago Ontañón. A closer look at invalid action masking in policy gradient algorithms. In *Proceedings of the International Florida Artificial Intelligence Research Society Conference (FLAIRS)*, 2022.

[5] David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and go through self-play. *Science*, 362(6419):1140–1144, 2018.

[6] Victor Allis. *Searching for Solutions in Games and Artificial Intelligence*. PhD thesis, Vrije Universiteit Amsterdam, 1994.

[7] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 278–287, 1999.